

Task 1 – Basic Network Scanning with Nmap

Cyber Security Internship

Prepared as an internship task submission

1. Objective

The purpose of this task was to get familiar with Nmap and use it for a basic TCP port scan. I wanted to see how a security analyst can identify which services are exposed on a host.

2. Tool and Target

Tool: Nmap 7.99.1 Target scanned: 192.168.34.238 The scan was performed on an authorized device/network.

3. Command Used

The command used to run the scan and save the output was: `nmap -oN "%USERPROFILE%\Downloads\scan_result.txt" 192.168.34.238`

4. What I Found

- 135/tcp was open and identified as msrpc.
- 139/tcp was open and identified as netbios-ssn.
- 445/tcp was open and identified as microsoft-ds.
- 1521/tcp was open and identified as oracle.
- 5500/tcp was open and identified as hotline.
- The host was up and Nmap reported 995 closed TCP ports.
- The scan covered one IP address and completed in about 0.89 seconds.

5. What the Results Mean

The scan gave me a quick view of the services that were reachable through TCP. An open port is not automatically a vulnerability, but it tells a security analyst where further investigation may be useful. Services such as SMB-related ports (139 and 445) and database-related port 1521 should be understood and secured according to the environment.

6. Problem I Faced

At first, Nmap could not save the output because I tried to write the file inside the Program Files directory. Windows denied access. I then changed the output location to my Downloads folder and the scan completed successfully.

7. What I Learned

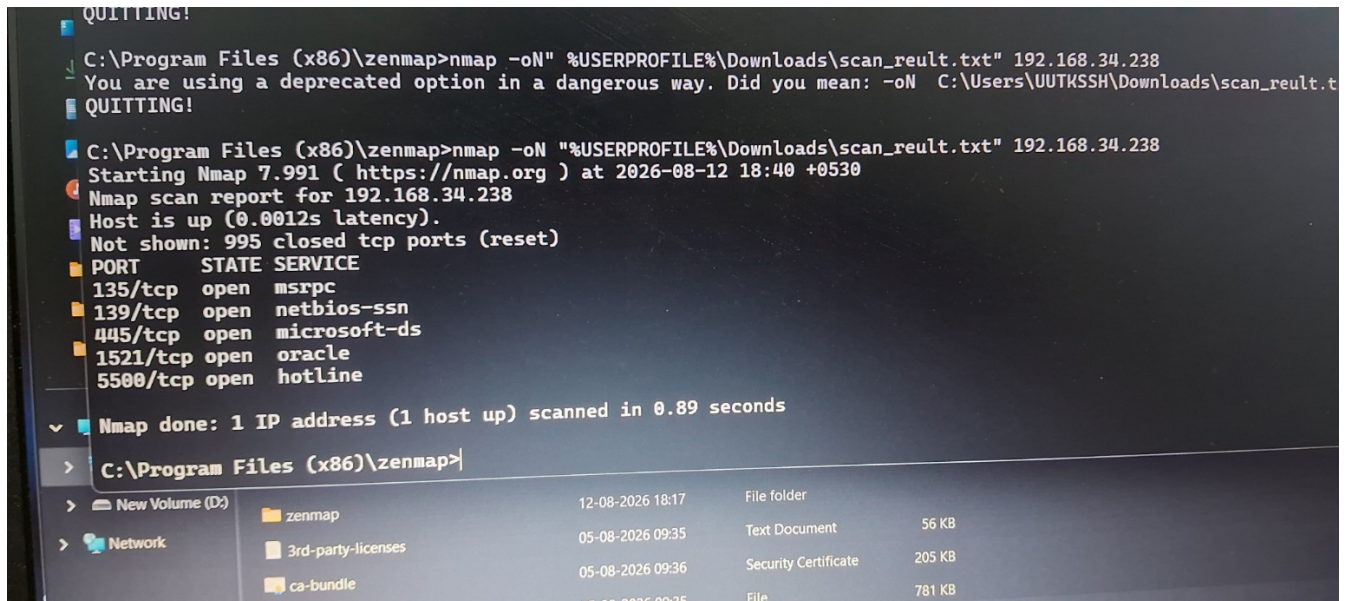
The main thing I learned from this task is that network scanning is a basic but important part of security assessment. I also learned how to save Nmap output so that the result can be reviewed later instead of relying only on the terminal screen.

8. Conclusion

The Nmap scan was completed successfully and the results were saved as scan_result.txt. This task gave me practical experience with basic port and service discovery.

Practical Evidence

The following screenshot shows the Nmap command, scan result, and the generated output file used for this task.



The screenshot displays a Windows command prompt window where an Nmap scan is executed. The command used is `C:\Program Files (x86)\zenmap>nmap -oN "%USERPROFILE%\Downloads\scan_result.txt" 192.168.34.238`. The output shows the scan starting at 2026-08-12 18:40 +0530, reporting that the host is up with a latency of 0.0012s, and listing several open ports: 135/tcp (msrpc), 139/tcp (netbios-ssn), 445/tcp (microsoft-ds), 1521/tcp (oracle), and 5500/tcp (hotline). Below the command prompt, a File Explorer window shows the contents of the 'Downloads' folder, including the generated output file 'scan_result.txt' (56 KB).

```
QUITTING!  
C:\Program Files (x86)\zenmap>nmap -oN "%USERPROFILE%\Downloads\scan_result.txt" 192.168.34.238  
You are using a deprecated option in a dangerous way. Did you mean: -oN C:\Users\UUTKSSH\Downloads\scan_result.t  
QUITTING!  
C:\Program Files (x86)\zenmap>nmap -oN "%USERPROFILE%\Downloads\scan_result.txt" 192.168.34.238  
Starting Nmap 7.991 ( https://nmap.org ) at 2026-08-12 18:40 +0530  
Nmap scan report for 192.168.34.238  
Host is up (0.0012s latency).  
Not shown: 995 closed tcp ports (reset)  
PORT      STATE SERVICE  
135/tcp    open  msrpc  
139/tcp    open  netbios-ssn  
445/tcp    open  microsoft-ds  
1521/tcp   open  oracle  
5500/tcp   open  hotline  
  
Nmap done: 1 IP address (1 host up) scanned in 0.89 seconds  
C:\Program Files (x86)\zenmap>  
  
New Volume (D:)  zenmap  12-08-2026 18:17  File folder  
Network          3rd-party-licenses  05-08-2026 09:35  Text Document  56 KB  
                  ca-bundle  05-08-2026 09:36  Security Certificate  205 KB  
                  05-08-2026 09:35  File  781 KB
```